

SAINT-GERMAIN-DE-GRANTHAM, QC, Canada

WMO: 719670

Lat: 45.8253N

Lon: 72.5369W

Elev: 282

StdP: 14.55

Time Zone: -5.00 (NAE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.8	-10.5	-22.0	1.6	-15.0	-16.9	2.2	-9.8	30.6	24.6	26.7	19.6	4.2	140	0.586

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.7	84.6	71.6	82.1	69.7	79.6	68.5	74.1	81.3	72.3	79.1	70.5	76.9	10.8	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.9	119.0	78.5	70.0	111.4	76.1	68.2	104.8	74.4	37.8	80.8	36.2	79.1	34.6	76.9	79.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
23.2	20.4	18.2	DB	-26.6	89.1	5.9	1.5	-30.9	90.2	-34.3	91.1	-37.7	91.9	-42.0	93.0	(4)
			WB	-26.8	77.0	5.9	1.9	-31.0	78.4	-34.5	79.5	-37.8	80.6	-42.1	82.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	42.9	13.9	16.6	26.5	41.7	55.7	62.9	68.3	66.1	58.9	46.9	34.4	21.7
	DBStd	21.23	13.51	12.20	11.98	8.21	7.80	6.34	5.33	5.09	7.77	8.27	9.56	12.85
	HDD50	4612	1118	935	732	273	31	1	0	0	11	161	472	878
	HDD65	8361	1583	1355	1194	700	304	114	26	50	213	562	918	1342
	CDD50	2029	0	0	3	23	206	388	566	498	278	64	3	0
	CDD65	306	0	0	0	0	14	50	127	83	31	1	0	0
	CDH74	2433	0	0	0	18	197	435	998	556	225	4	0	0
	CDH80	465	0	0	0	2	40	88	224	81	30	0	0	0
Wind	WSAvg	8.2	9.3	9.5	9.2	9.9	8.6	7.5	6.7	6.2	6.4	7.8	8.6	8.5
Precipitation	PrecAvg	42.6	3.2	2.3	2.7	3.2	3.5	4.1	4.5	4.3	3.8	4.1	3.5	3.5
	PrecMax	52.2	4.7	3.7	4.0	5.8	6.8	6.5	7.6	6.7	7.3	8.6	5.7	6.9
	PrecMin	33.2	1.5	1.4	1.1	0.3	1.9	1.7	2.4	2.2	2.0	1.5	1.3	1.4
	PrecStd	5.1	1.0	0.6	0.7	1.5	1.3	1.4	1.4	1.2	1.3	1.8	0.9	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.2	44.3	63.1	76.3	84.4	86.7	88.9	85.3	83.7	73.4	62.7	50.2
		MCWB	44.8	41.1	57.9	62.8	66.9	70.3	75.6	72.5	73.5	61.7	55.0	47.0
	2%	DB	40.1	40.0	50.6	67.4	79.5	82.6	85.2	82.2	79.7	68.7	56.2	45.0
		MCWB	38.2	37.0	43.9	56.1	65.0	68.2	73.0	70.7	69.2	61.1	51.0	43.5
	5%	DB	36.7	36.9	45.0	60.6	74.9	79.2	82.4	79.7	76.3	64.4	52.0	40.9
		MCWB	34.8	34.3	40.2	51.2	61.4	66.2	70.8	69.2	66.9	57.6	48.2	39.3
	10%	DB	33.2	34.6	41.4	55.6	70.8	75.5	79.6	77.0	72.8	60.1	48.2	37.0
		MCWB	31.6	32.8	37.6	47.8	59.8	64.5	69.5	67.8	65.5	54.7	44.8	35.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.2	42.3	57.9	63.2	71.3	74.1	77.7	74.8	74.2	66.8	56.6	47.4
		MCDB	46.0	43.8	62.7	74.6	81.2	82.4	86.5	82.7	81.5	69.8	59.8	49.6
	2%	WB	38.6	37.1	44.8	58.2	67.5	71.2	75.1	72.6	71.2	62.5	52.0	43.4
		MCDB	39.3	39.0	49.1	64.2	75.3	79.1	82.1	79.5	76.7	67.7	55.0	44.9
	5%	WB	34.5	34.7	41.0	52.5	64.8	68.6	72.9	70.8	68.8	58.5	48.2	39.3
		MCDB	36.3	36.5	44.0	58.4	71.0	75.2	79.3	77.1	74.2	63.0	51.1	40.8
	10%	WB	31.9	32.9	37.6	48.8	61.7	66.5	71.2	69.2	66.6	55.5	44.8	35.5
		MCDB	33.1	34.6	40.7	55.3	68.5	72.8	77.7	75.3	71.8	59.2	48.1	36.6
Mean Daily Temperature Range	5% DB	MDBR	17.5	18.0	17.0	17.6	21.0	19.6	19.7	19.8	20.8	17.0	14.3	14.4
		MCDBR	16.7	17.0	19.6	28.0	27.9	24.1	21.8	22.1	23.0	23.1	20.3	14.9
		MCWBR	16.4	15.6	14.9	18.2	14.9	12.3	11.6	12.3	13.5	15.0	15.7	14.5
	5% WB	MCDBR	16.2	16.0	18.0	24.8	22.2	19.2	18.8	19.0	20.0	19.2	19.2	14.9
		MCWBR	16.0	15.0	14.4	17.6	14.4	11.3	11.1	11.3	13.0	14.1	15.8	14.7
Clear-Sky Solar Irradiance	taub	0.257	0.288	0.322	0.354	0.379	0.419	0.423	0.407	0.373	0.330	0.307	0.284	
	taud	2.363	2.254	2.304	2.369	2.357	2.244	2.236	2.283	2.372	2.495	2.477	2.348	
	Ebn at Noon	273	282	287	287	282	270	267	266	267	266	251	246	
	Edh at Noon	23	31	35	36	38	43	43	39	33	25	21	21	
All-Sky Solar Radiation	RadAvg	456	751	1149	1397	1625	1784	1781	1570	1239	718	432	313	
	RadStd	55	80	97	155	159	133	141	73	101	61	56	26	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (47 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page